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Development and Spatial Validation of a Random Forest Prediction Model for Firearm-Related Injury Risk in Chicago Census Tracts

Gia Barboza-Salerno, MA, MS, JD, PhD[1,2]

Hexin Yang, MS[1]

Taylor Harrington, MS[2]

The Ohio State University

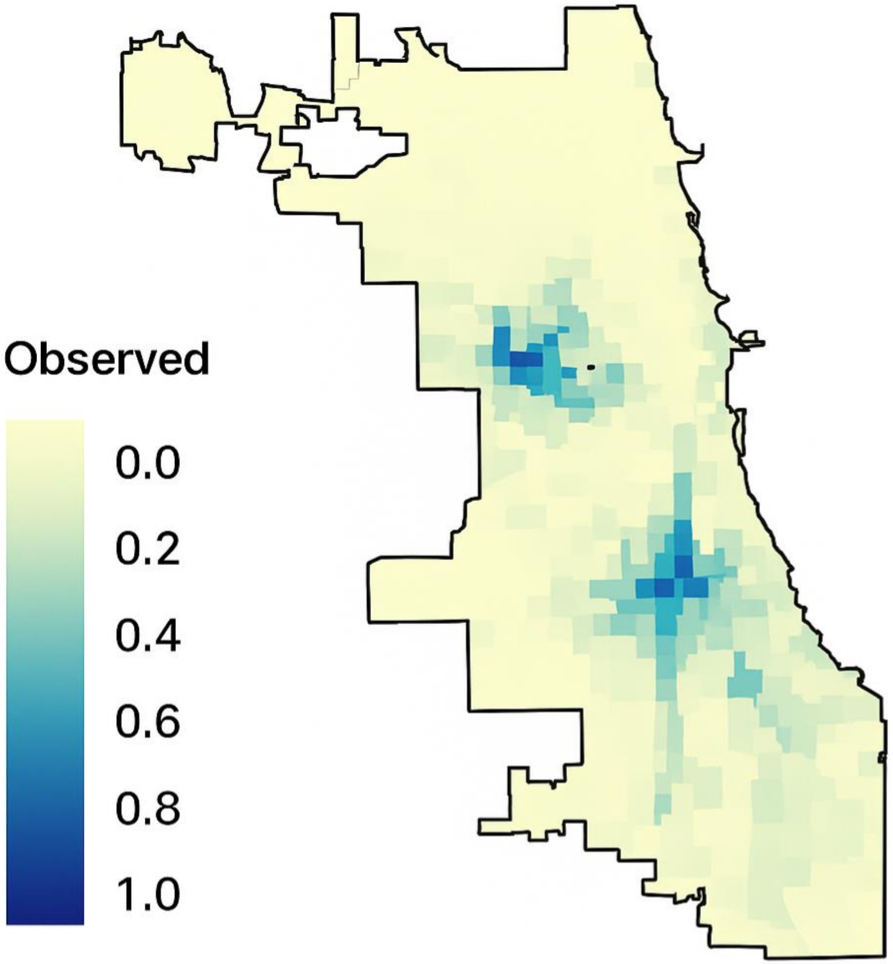
[1]College of Social Work

[2]College of Public Health

INVESTIGATING SPATIAL STRUCTURES in Urban Environments FOR LEGAL AND POLICY INSIGHTS
LAB

Background

- Background
  - Firearm injury is a ***predictable and preventable*** public health crisis, particularly in densely populated urban areas
  - Between 2018–2023:
    - 1,274 firearm homicides under age 25, 97.9% of victims are non-Hispanic Black
    - 65% of all deaths among Black males aged 18–24
    - Disproportionate impact on low-income communities of color
  - Reflects entrenched racial, socioeconomic, and ***geographic*** inequities

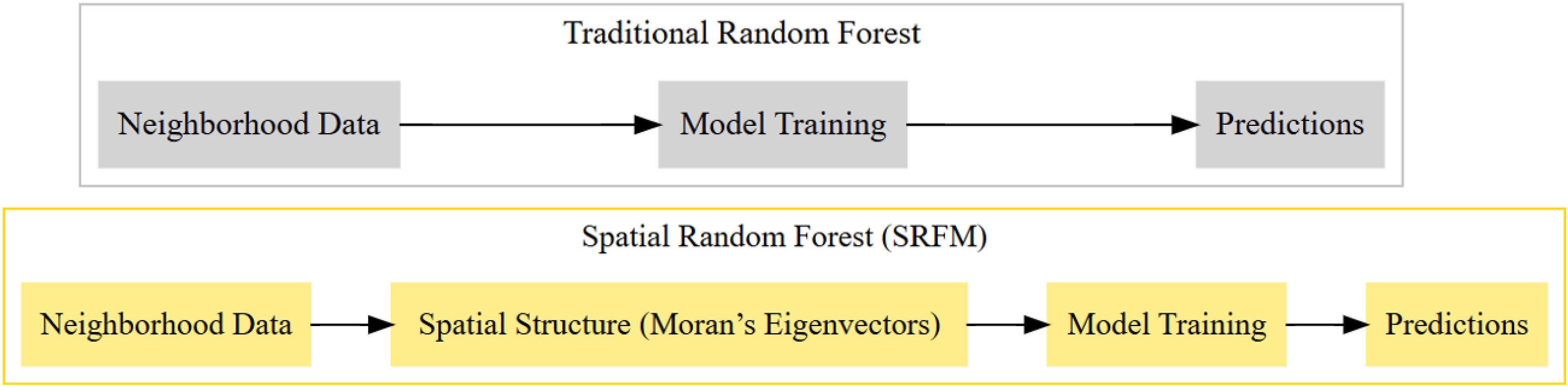


Structural and Environmental Context

- Violence concentrated in areas of racial segregation, economic deprivation, and environmental injustice
- Protective environmental features: green space, trees, parks → associated with reduced violence and improved mental health (Hartley et al., 2021; Liu et al., 2023)
- Despite repeated calls for prevention-oriented, structural interventions, research remains overly focused on individual or justice outcomes

Research Gaps

- Non-spatial models (RFMs, GAMs) ignore spatial dependence → inflated variable importance, biased inference
- Focus on homicides — ignores nonfatal shootings (2–3× more frequent)
- Limited integration of social, built, and natural environment predictors



Study Aim and Rationale

- This study applies Spatial Random Forest Models (SRFMs) to predict non-fatal firearm injuries (2020–2024) in Chicago census tracts, addressing the limitations of traditional models by explicitly modeling spatial dependence
- Goals:
  - Compare predictive accuracy of SRFM vs NSRFM
  - Test if spatial models eliminate residual spatial autocorrelation
  - Identify the predictors of firearm injury risk through a place-based framework that recognizes violence as shaped by the social, built, and natural environment

Data Source	Description
Chicago Police Department	Publicly available incident-level dataset documenting non-fatal firearm shootings, including date, location, and incident characteristics used to aggregate tract-level firearm injury rates. https://data.cityofchicago.org/Public-Safety/Violence-Reduction-Dataset/
American Community Survey	Provides tract-level demographic, socioeconomic, and housing data including population counts, racial composition, income, education, and employment indicators. https://www.census.gov/programs-surveys/acs
Healthy Regions & Policies Lab	A harmonized Chicago Environmental and Social Vulnerability dataset integrating environmental exposures (lead, PM2.5, flood risk), health outcomes, and social vulnerability indices. https://chichives.com
Capital Flows in Chicago	Dataset capturing tract-level public and private capital investments across asset classes, including residential, commercial, industrial, and multifamily properties. https://datacatalog.urban.org/dataset/exploring-capital-flows-chicago
Home Mortgage Disclosure Act	Federal dataset documenting mortgage originations, applications, and denials by race, ethnicity, and tract, used to assess racialized homeownership and lending disparities. https://ffiec.cfpb.gov/data-publication/modified-lar
Business Licensing Database	Official municipal licensing data identifying the location and type of alcohol outlets used to calculate tract-level alcohol outlet density. https://data.cityofchicago.org/Business/Business-Licenses/

Measures

- Dependent Variable
  - Non-fatal shooting rate per 1,000 residents, aggregated to census tracts
- Key Predictors (37 total)
  - *Natural*: Tree canopy change, NDVI, urban heat, flood risk
  - *Built*: Traffic volume, alcohol outlets, housing cost burden, job accessibility
  - *Social/Structural*: Race, SVI, hardship, investment, income share, homeownership disparity

Modeling Framework

- Non-Spatial RF (NSRFM)
  - Standard random forest including coordinates.
- Spatial RF (SRFM)
  - Incorporates Moran’s Eigenvector Maps (MEMs) from tract centroid distances to capture geographic dependence.
- Cross-validation
  - 30-fold *spatial* (via k-means on centroids)
  - R^2, pseudo-R^2, RMSE, Moran’s I.

Table 1

Comparison of Spatial and Non-Spatial Random Forest Model Metrics

Metric	Non-spatial RF	Spatial RF
Number of predictors	17	46 (including spatial MEMs)
OOB R^2	0.6905	0.6881
True R^2	0.9559	0.9601
Pseudo R^2	0.9777	0.9799
OOB RMSE	11.9913	12.0378
RMSE	5.1287	5.0185
Normalized RMSE	0.223	0.2182
Residual Spatial Autocorrelation	Significant positive autocorrelation	No spatial autocorrelation
Residual Normality (Shapiro-Wilk W)	0.817 (not normal)	0.800 (not normal)

Table 2

Variable Importance from Spatial Random Forest Model

Description	Importance Score
Percent of population identifying as Black	11.25
Percent of population identifying as White	10.70
Percent of children aged 1-5 with elevated blood lead levels	7.969
CDC Social Vulnerability Index score	7.332
Tree canopy change from 2010 to 2017	6.791
Percent of foreign-born residents	5.950
Economic hardship index based on ACS indicators	5.887
Street-level traffic volume score	5.572
Median income per tract relative to entire MSA median	5.352
Percent of investment in single-family housing	5.275
Median gross rent (2018)	4.971
Percent of households spending >30% of income on housing	4.671
White homeownership share divided by household share	4.279
Black homeownership share divided by household share	4.177
Land area of census tract in square miles	4.169
Jobs accessible within 45-minute transit commute	4.068

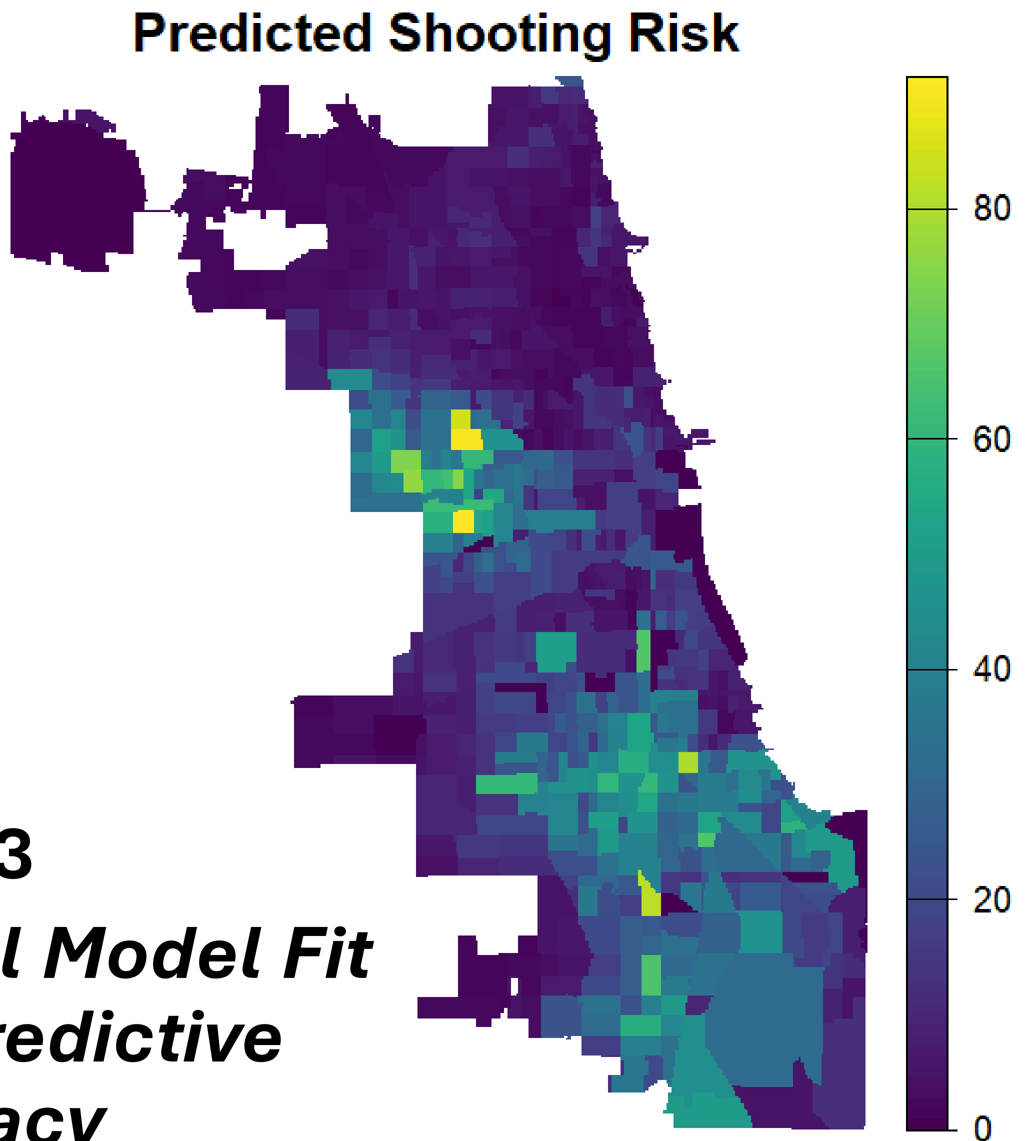
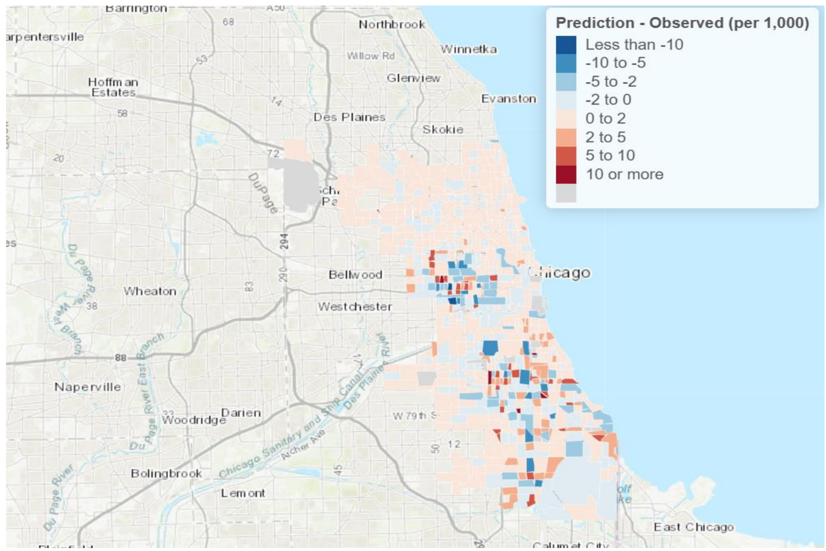


Fig. 1-3

Spatial Model Fit and Predictive Accuracy



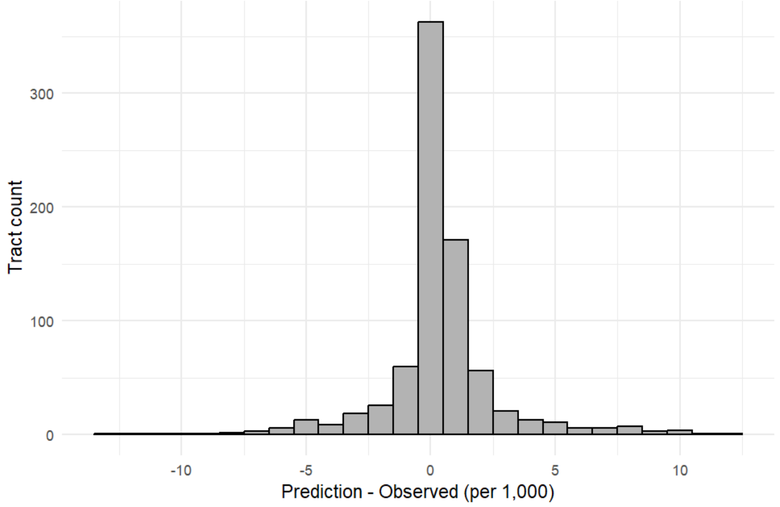


Fig. 4

Local Variable Importance

- Spatial Random Forest models capture geographic heterogeneity in predictor strength
  - Race and social vulnerability operate city-wide
  - Tree canopy, rent burden, and job access have localized, context-specific influences

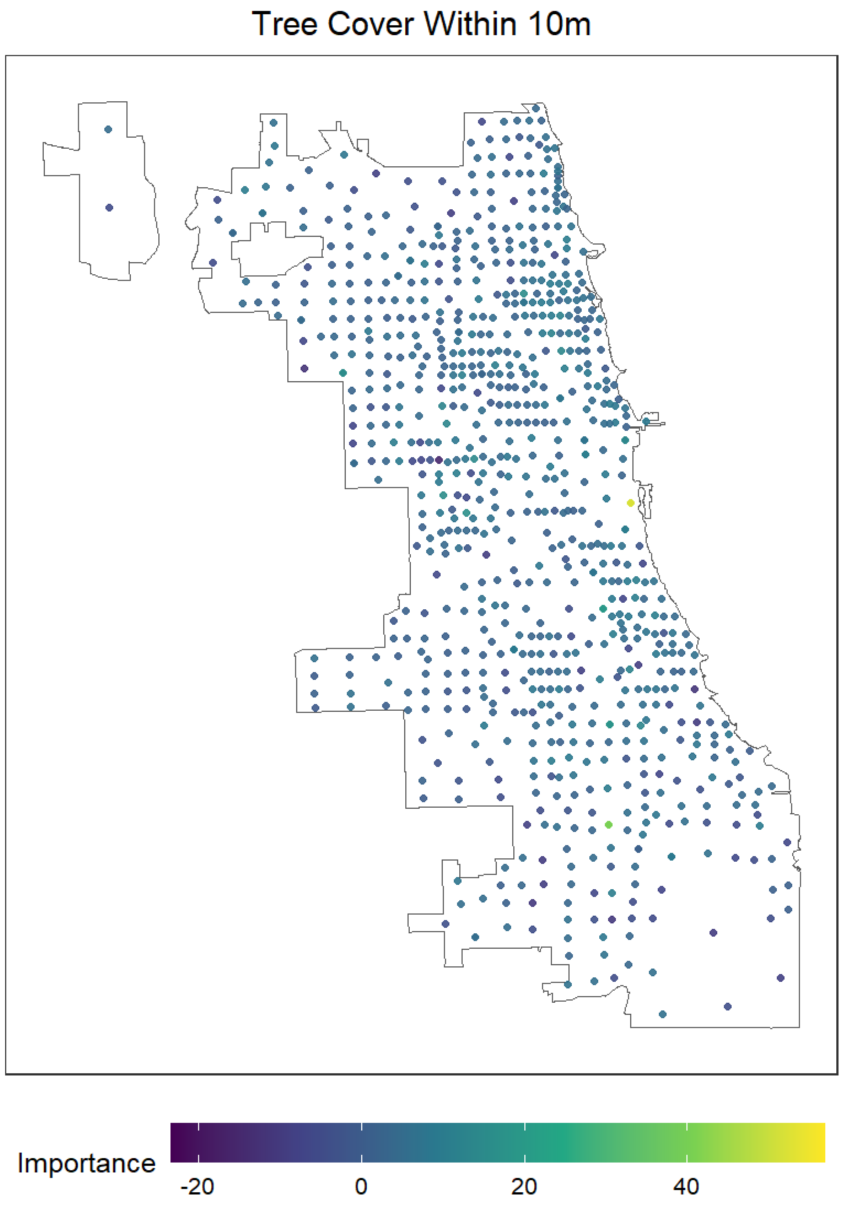


Fig. 4

Partial Dependence Plots: Threshold Effects

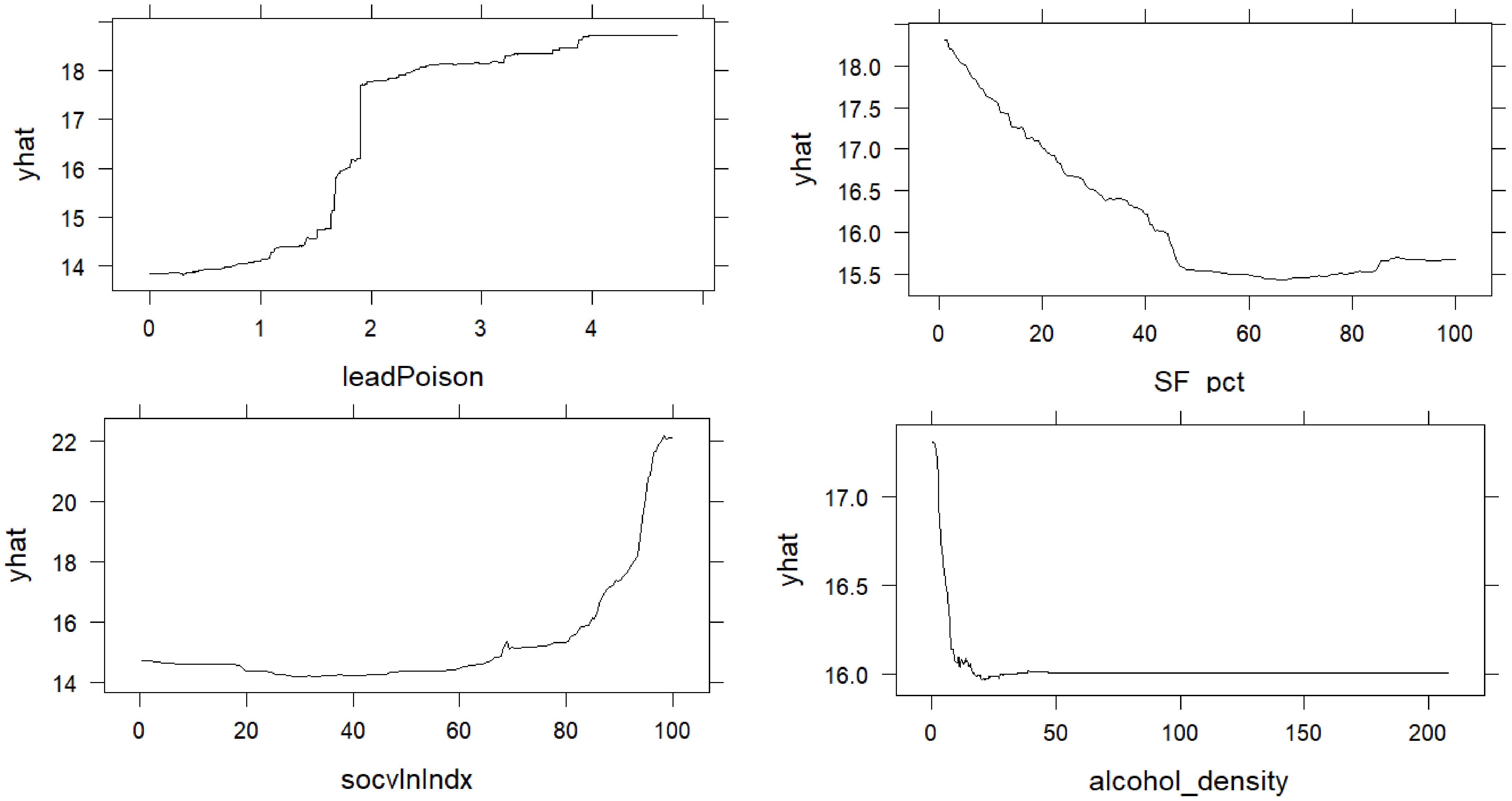


Fig. 5

Partial Dependence Plots: Conditional Effects

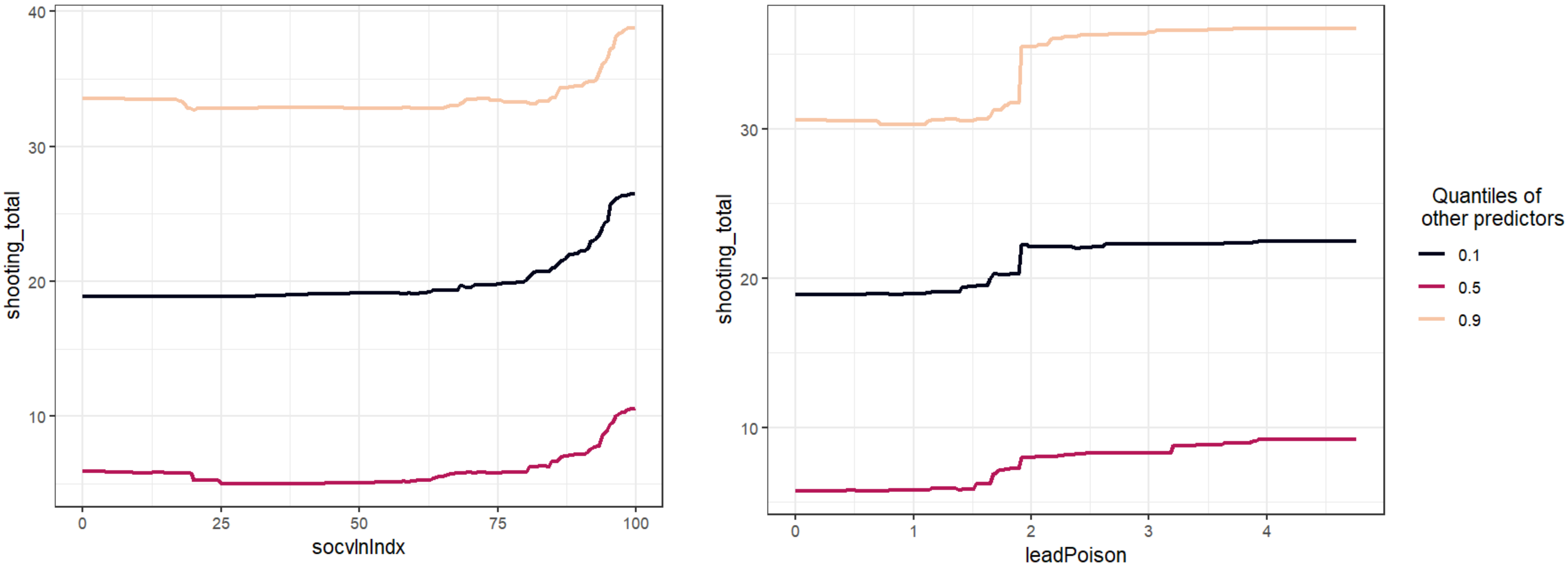
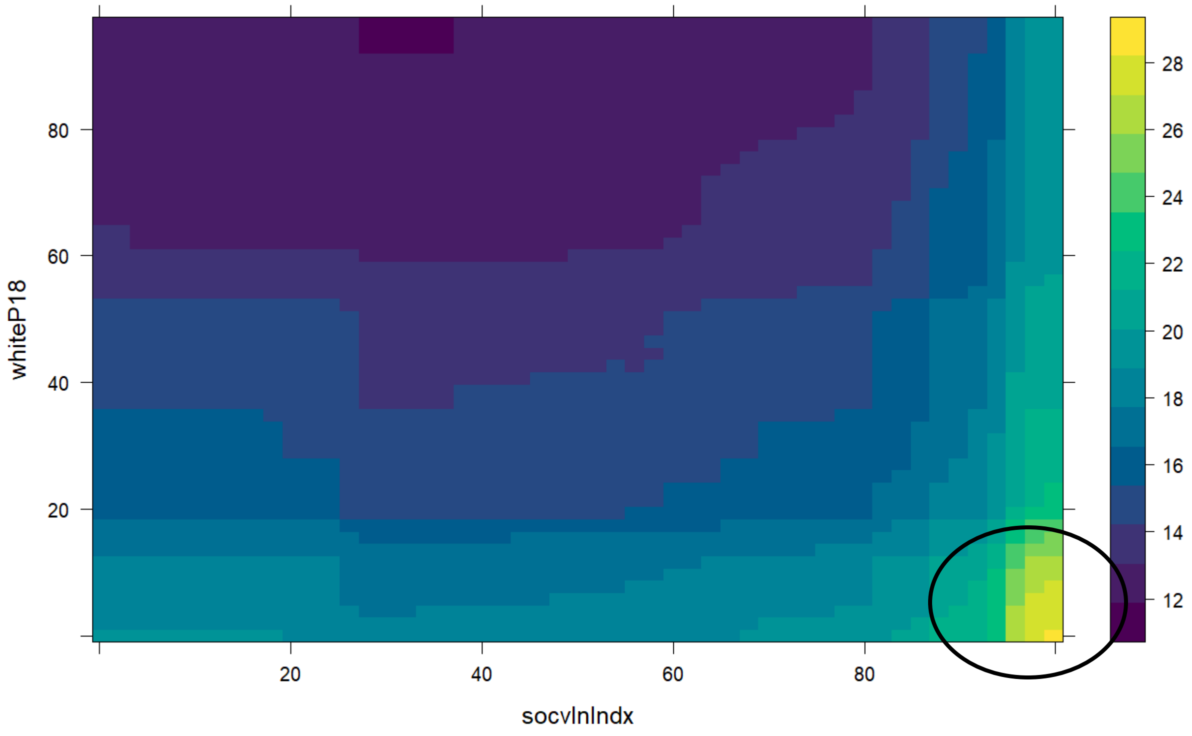


Fig. 4

Partial Dependence Plots: Interaction Effects



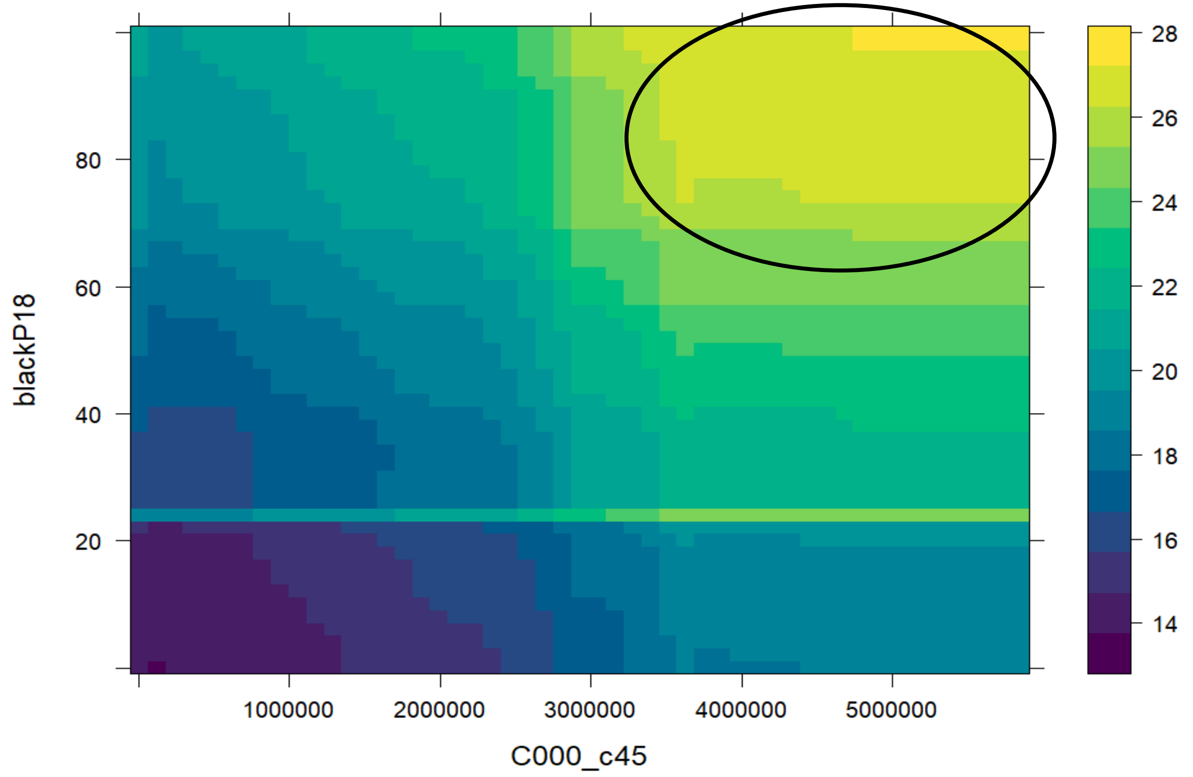


Table 3

Key Predictor Distributions by Firearm Violence Quintile

Variable	Q1	Q2	Q3	Q4	Q5
Percent Black	5.99 (9.15)	11.47 (18.70)	17.44 (26.72)	59.99 (38.76)	85.86 (19.61)
Percent White	75.29 (18.15)	66.66 (20.03)	53.81 (22.84)	23.18 (24.38)	8.04 (13.09)
Lead Poisoning Rate	0.69 (0.64)	0.99 (0.72)	1.22 (0.73)	1.98 (0.95)	2.77 (0.98)
Tree Canopy Change	0.66 (0.91)	0.66 (1.04)	0.64 (1.00)	0.69 (1.17)	0.72 (0.96)
Percent Foreign-Born	19.74 (13.77)	23.42 (12.55)	27.51 (13.54)	14.62 (15.37)	4.69 (7.01)
Economic Hardship	14.62 (6.96)	19.06 (8.08)	24.84 (10.16)	28.07 (9.22)	35.85 (10.41)
Traffic Volume	4.66 (1.66)	4.70 (1.52)	4.76 (1.28)	4.84 (1.02)	4.96 (0.96)
Median Income share	140.39 (64.17)	110.31 (57.59)	82.87 (48.50)	64.13 (41.24)	53.11 (24.86)
Single Family Housing	68.41 (20.25)	65.84 (22.93)	52.05 (26.59)	34.04 (25.55)	29.20 (21.75)
Land Area Sq.Mi.	0.34 (0.67)	0.24 (0.19)	0.26 (0.32)	0.27 (0.20)	0.33 (0.58)
Job Accessibility	971,472.60 (1,070,200.30)	1,081,070.60 (1,019,788.80)	902,777.80 (919,404.90)	652,674.10 (778,630.20)	721,681.50 (766,807.60)

Table 4

Sensitivity Analysis

Definition of High-Risk Area	Threshold (Predicted/1,000)	Sensitivity	Specificity
Threshold-Based: Predicted ≥ 2 per 1,000	2	1.0000	0.4907
Threshold-Based: Predicted ≥ 4 per 1,000	4	1.0000	0.7050
Threshold-Based: Predicted ≥ 6 per 1,000	6	1.0000	0.7888
Threshold-Based: Predicted ≥ 8 per 1,000	8	1.0000	0.8696
Threshold-Based: Predicted ≥ 10 per 1,000	10	0.9938	0.9286
Confusion Matrix			
60th Percentile	-	0.9536	0.9689
80th Percentile	-	0.9012	0.9752

Conclusion

- Summary
  - Study aim and innovation
  - Superior model performance
  - Comprehensive integration of social, natural, and built environment predictors
  - Key structural determinants identified
- Policy implications
  - Shift Focus to Structural and Preventive Interventions
  - Target Core Structural Determinants Identified by the Model
  - Implement Spatially Differentiated Strategies
  - Utilize the SRFM for Proactive Resource Allocation